Filing Rate Decomposition and Back-Rent Analysis

Philadelphia Evictions, 2011–2019

2026-03-26

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1 Overview

This report summarizes two analyses of the Philadelphia eviction record (2011–2019):

1. **Entity consolidation** (`build-ownership-panel.R`): A two-phase owner linkage pipeline mapping RTT transfers and OPA parcels to legal entities and conglomerates.
2. **Filing rate decomposition** (`analyze-filing-decomposition.R`):
   - *Part A*: What fraction of cross-sectional variation in building-level filing rates is attributable to building characteristics, neighborhood, tenant composition, and landlord type?
   - *Part B*: Do corporate or high-filing landlords file evictions with less accumulated back rent? Is the plaintiff (often a management company) different from the owner?

Commercial case filter (Part B): Commercial cases are identified by the defendant in position 1 of the party-names table, not by the self-reported `commercial` flag in the evictions data (which is unreliable). Counterclaim parties (e.g. a corporate landlord counter-sued by a tenant) are excluded from the defendant classification. This is produced by `impute-race-preflight.R` (`eviction_case_commercial` product).

Sample restrictions (Part B): non-commercial cases, `ongoing_rent` 300–6000, `total_rent` 300–15000, `months >= 1`, single-parcel PID match, 2011–2019.

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2 Part 0: Entity Consolidation

2.1 Two-phase owner linkage

Phase 1 normalizes raw RTT grantee names (stripping entity suffixes, roman numerals, bank/trust fragments) and groups exact-match normalized names into **entities**. Persons and trusts are grouped by *(name, PID)* rather than name alone, preventing common names from forming spurious hubs.

Phase 2 uses a name+address self-join on OPA mailing data: corporations match on *(business_short_name, address)* where `business_short_name` further strips functional descriptor words (PROPERTIES, MANAGEMENT, REALTY, etc.); persons/trusts match on *(exact name, address)*. A registered-agent filter drops addresses linked to more than 300 distinct entities. Connected components of the resulting name+address graph form **conglomerates**.

Phase 3 adds one-shot RTT→OPA transitive edges (last RTT grantee vs. 2024 OPA owner for each PID), subject to a name-similarity gate: corporations must share the same `business_short_name` (4 chars); persons must share 1 token of 5 characters. This prevents government entities (City of Philadelphia, PHA) from absorbing hundreds of unrelated parcels acquired by tax-exempt conveyances not captured in RTT.

2.2 PID × year coverage

2.3 Largest entities and conglomerates by category

Table 1: Entity consolidation summary

Stage	Count
Phase 1 entities (RTT exact-name match)	451,346
OPA-only entity additions	262,935
Total entities	714,281
Registered-agent addresses dropped (>300 entities each)	44 (all SIMPLIFILE LC E-RECORDING variants)
Entities entering mailing-address graph	64,461
Conglomerates formed	49,651
Multi-entity conglomerates	3,063
Phase 1 entities merged by Phase 2	4,464
Name lookup entries (all sources)	719,215

Table 2: PID x year ownership panel coverage

Metric	Value
PID x year panel rows	14,015,256 (583,969 PIDs x 24 years)
RTT-sourced owner rows	5,128,881 (36.6%)
OPA-fallback owner rows	8,886,375 (63.4%)
Entity ID coverage	14,015,256 (100%)
Portfolio bin: 1 PID	3,522,595 (43.9% of 2011-19 PID-yrs)
Portfolio bin: 2–4 PIDs	981,376 (12.2%)
Portfolio bin: 5–19 PIDs	425,827 (5.3%)
Portfolio bin: 20+ PIDs	325,923 (4.1%)

Table 3: Largest sample entities by owner category (top two per category).

Category	Rank	Primary name	Entity ID	Max PIDs	Max units	Sample PIDs	Entities in CG
Financial-Intermediary	1	FEDERAL NATIONAL MORTGAGE ASSOCIATION	117	343	416	1,285	1
Financial-Intermediary	2	WELLS FARGO BANK NA	1544	225	260	428	1
For-profit corp	1	WEST VILLAGE GROUP LLC	15168	87	755	87	2
For-profit corp	2	VDBE WOODCREST LLC	50564	20	696	20	1
Government-Federal	1	UNITED STATES OF AMERICA INC	576683	26	53	26	1
Government-Federal	2	ADMIN OF VETERAN AFFAIRS	580202	11	11	11	1
Government-Local	1	CITY OF PHILA	566959	1,655	1,215	1,655	293
Government-Local	2	SEPTA	810245	278	820	278	1
Nonprofit	1	TEMPLE UNIV COMMONWEALTH SYSTEM OF HIGHER ED	576224	112	995	112	1
Nonprofit	2	GEDR UPENN LLC	576729	146	757	146	42
PHA	1	HLP PHILADELPHIA HOUSING	4796	5,543	9,883	5,544	30
PHA	2	PHILADELPHIA HOUSING AUTHORITY DEVELOPMENT CORPORATION	22597	4	61	4	30
Person	1	DOMB ALLAN	140314	226	548	226	34
Person	2	SUMMIT PARK EAST	825045	2	487	2	1
Religious	1	EVANGELICAL MANOR	654110	1	159	1	1
Religious	2	TIOGA PRESBYTERIAN APARTMENTS INC	576320	3	116	3	1
Trust	1	BIENENFELD JACK TR JACK BIENENFELD SEPARATE SHARE TRUST UND	565889	2	108	2	1
Trust	2	2011 IRREVOCABLE TRUST FOR FAMILY OF HEN BIENENFELD JACK TR	562833	1	74	1	1

Table 4: Largest sample conglomerates by primary owner category (top two per category).

Category	Rank	Primary name	CG ID	Primary entity ID	Entities in CG	Max PIDs	Max units
Financial-Intermediary	1	FEDERAL NATIONAL MORTGAGE ASSOCIATION	117	117	1	343	416
Financial-Intermediary	2	WELLS FARGO BANK NA	1544	1544	1	225	260
For-profit corp	1	PARK BRADFORD APARTMENTS	1028	573283	17	72	1,011
For-profit corp	2	FERNROCK APARTMENTS 2 LP	3203	568708	53	206	983
Government-Federal	1	UNITED STATES OF AMERICA INC	576683	576683	1	26	53
Government-Federal	2	ADMIN OF VETERAN AFFAIRS	580202	580202	1	11	11
Government-Local	1	CITY OF PHILA	1	566959	293	5,496	2,700
Government-Local	2	SEPTA	810245	810245	1	278	820
Nonprofit	1	GEDR UPENN LLC	1303	576729	42	255	1,586
Nonprofit	2	TEMPLE UNIV COMMONWEALTH SYSTEM OF HIGHER ED	576224	576224	1	112	995
PHA	1	HLP PHILADELPHIA HOUSING	4796	4796	30	7,462	10,384
PHA	2	PHILADELPHIA HOUSING ASSOCIATES LLC	6718	6718	1	3	6
Person	1	DOMB ALLAN	986	140314	34	302	929
Person	2	SUMMIT PARK EAST	825045	825045	1	2	487
Religious	1	TEMPLE VILLAS II LLC	13128	15134	39	67	242
Religious	2	EVANGELICAL MANOR	654110	654110	1	1	159
Trust	1	BIENENFELD JACK TR JACK BIENENFELD SEPARATE SHARE TRUST UND	565889	565889	1	2	108
Trust	2	2011 IRREVOCABLE TRUST FOR FAMILY OF HEN BIENENFELD JACK TR	562833	562833	1	1	74

3 Part A: Cross-Sectional Filing Rate Decomposition

Sample: ever-rental buildings in 2011–2019 with non-missing EB filing rate, zip, census tract, and `rental_share`. N reported in table below. No minimum unit count restriction — if a building appears in eviction filings it is treated as a rental.

Rental share correction: The EB filing rate denominator includes all observed unit-years, but single-family homes are sometimes rented only intermittently, deflating their apparent filing rates. All models (M1 onward) include `rental_share` = fraction of sample years with an active rental license, correcting for this. Philadelphia Housing Authority (PHA) buildings are exempt from the City licensing requirement; their `rental_share` is imputed to 1 (always fully rented).

3.1 A1: Descriptive means by owner type and portfolio

Nonprofits stand out with a mean EB rate of 0.18 — far above the other types and likely reflecting a small set of high-density affordable-housing buildings with concentrated poverty. PHA-owned buildings also exceed person-owned buildings (0.10 vs. 0.05). For-profit corporations file at 0.07, modestly above the 0.05 person baseline.

Table 5: EB filing rate by owner type and portfolio (cross-section, 2011–2019). Mean units owned = avg total units across all properties in the landlord's portfolio. Portfolio rows exclude PHA and government owners.

Group	N (PIDs)	Mean units owned	Mean EB rate	Median EB rate
Owner type (Person = reference)				
Owner: Person	140,388	4	0.040	0.005
Owner: For-profit corp	40,023	26	0.064	0.005
Owner: Nonprofit	1,110	110	0.170	0.131
Owner: PHA	5,471	9,593	0.011	0.005
Owner: Religious	425	15	0.035	0.005
Owner: Trust	810	6	0.061	0.005
Owner: Government-Local	372	384	0.034	0.005
Owner: Government-Federal	6	29	0.007	0.004
Owner: Financial-Intermediary	949	46	0.055	0.005
Entity portfolio (non-PHA / non-government)				
Entity: 1 prop.	139,913	3	0.041	0.005
Entity: 2-4 prop.	21,451	7	0.051	0.005
Entity: 5-19 prop.	13,730	16	0.068	0.005
Entity: 20+ prop.	8,611	112	0.086	0.008
Conglomerate portfolio (non-PHA / non-government)				
Cong.: 1 prop.	123,689	4	0.039	0.005
Cong.: 2-4 prop.	28,771	6	0.048	0.005
Cong.: 5-19 prop.	17,762	17	0.064	0.005
Cong.: 20+ prop.	13,483	183	0.086	0.007

Table 6: Race-augmented version of Table 5. Building Black share is unit-weighted across buildings; case Black probability is case-weighted across linked non-commercial cases with race imputation. Portfolio rows exclude PHA and government owners.

Group	N (PIDs)	Mean units owned	Mean EB rate	Mean bldg. Black share	Mean case Black prob.
Owner type (Person = reference)					
Owner: Person	140,388	4	0.040	32.7%	54.2%
Owner: For-profit corp	40,023	26	0.064	33.5%	56.0%
Owner: Nonprofit	1,110	110	0.170	53.9%	76.8%
Owner: PHA	5,471	9,593	0.011	73.9%	79.1%
Owner: Religious	425	15	0.035	46.9%	81.0%
Owner: Trust	810	6	0.061	30.4%	44.7%
Owner: Government-Local	372	384	0.034	17.0%	45.3%
Owner: Government-Federal	6	29	0.007	41.4%	2.7%
Owner: Financial-Intermediary	949	46	0.055	47.7%	63.5%
Entity portfolio (non-PHA / non-government)					
Entity: 1 prop.	139,913	3	0.041	31.2%	52.1%
Entity: 2-4 prop.	21,451	7	0.051	34.5%	59.9%
Entity: 5-19 prop.	13,730	16	0.068	43.6%	61.0%
Entity: 20+ prop.	8,611	112	0.086	52.7%	74.2%
Conglomerate portfolio (non-PHA / non-government)					
Cong.: 1 prop.	123,689	4	0.039	31.2%	52.4%
Cong.: 2-4 prop.	28,771	6	0.048	33.1%	55.2%
Cong.: 5-19 prop.	17,762	17	0.064	39.1%	57.5%
Cong.: 20+ prop.	13,483	183	0.086	43.9%	66.9%

3.2 A1b: Owner-Type Bridge

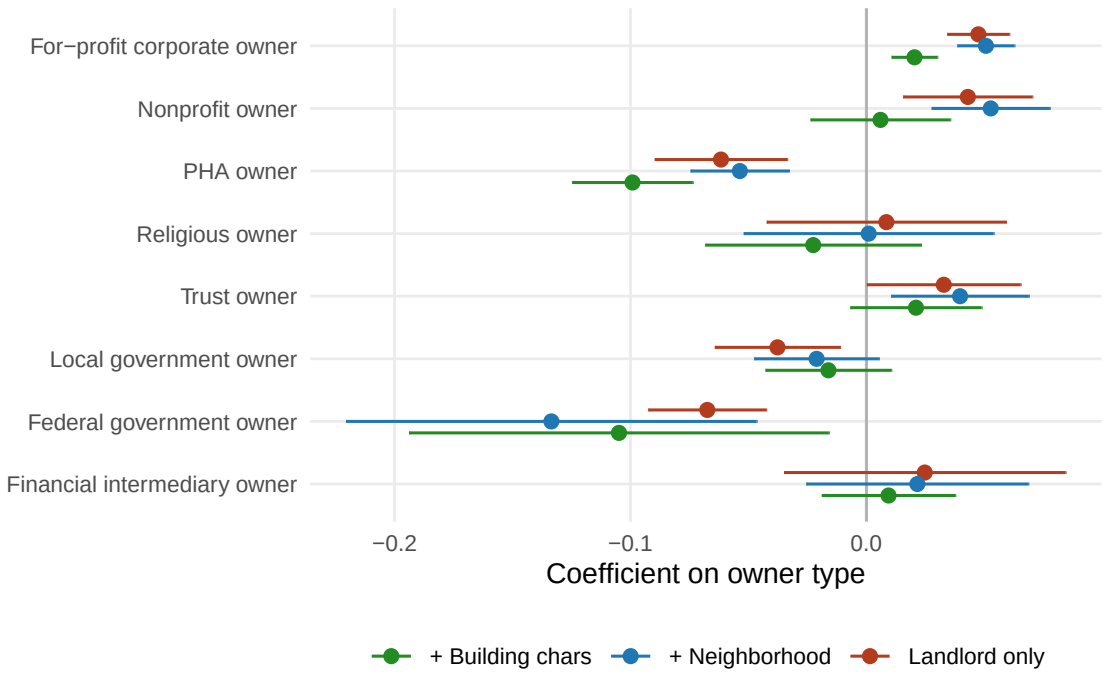

Figure 1: Owner-type coefficients as controls are added. Reference group: person owners. Error bars show 95% confidence intervals.

The nonprofit coefficient falls sharply as neighborhood and then building controls enter. That is the core reason the unconditional mean in Table 5 is so much larger than the conditional owner-type coefficient in Table 7: nonprofits disproportionately own larger conglomerate portfolios and large multifamily buildings in higher-filing tracts, but once those comparisons are made within tract/building cells, the residual nonprofit-owner effect is small.

3.3 A2: Sequential R^2 Decomposition

Six nested OLS models, weighted by `total_units`, outcome = `filing_rate_eb_pre_covid`. ΔR^2 isolates the variance attributable to each group of factors added. M4 and M5 both branch from M3: M4 uses entity-level portfolio (per LLC/name); M5 uses conglomerate-level portfolio (across all linked LLCs under common ownership, from improved Phase 2/3 reconciliation). M5's ΔR^2 is relative to M3.

Interpretation. Geography dominates the decomposition. The ΔR^2 for the ZIP→tract upgrade alone (M2→M3) is roughly 9× the landlord component ($\Delta R^2 \approx 0.061$ vs 0.007). The total geographic contribution — from no geographic controls (M0) through building characteristics (M1) to ZIP FE (M2) to tract FE (M3) — is substantially larger still: $\Delta R^2 \approx 0.208$ for all geographic FE vs ≈ 0.007 for landlord type, a roughly 30× ratio. Tenant racial composition adds a further increment beyond tract FE, suggesting neighborhood sorting by race is a distinct channel from simple geographic sorting. The `rental_share` covariate (included from M1) corrects for the deflated

Table 7: Sequential R^2 decomposition. Between-neighborhood raw variance share: 0.118.

Model	Description	R^2	Adj. R^2	ΔR^2	N
M0	Intercept only	—	—	—	188,435
M1	Zip FE + building chars + rental share	0.1467	0.1463	—	188,426
M2	Tract FE + building chars + rental share	0.2081	0.2063	0.0614	188,432
M3	Tract FE + building + pct_black + rental share	0.2190	0.2171	0.0109	174,904
M4	Tract FE + building + pct_black + rental share + landlord (entity portfolio)	0.2300	0.2281	0.0110	174,904
M5	Tract FE + building + pct_black + rental share + landlord (conglomerate portfolio)	0.2312	0.2292	0.0121	174,904

Note: Weighted by mean total units. Cluster SE at tract.

filing rates of intermittently-rented properties; its ΔR^2 contribution is absorbed into M1 and is not separately isolated. The landlord-type effect is small but present; numbers updated after re-run.

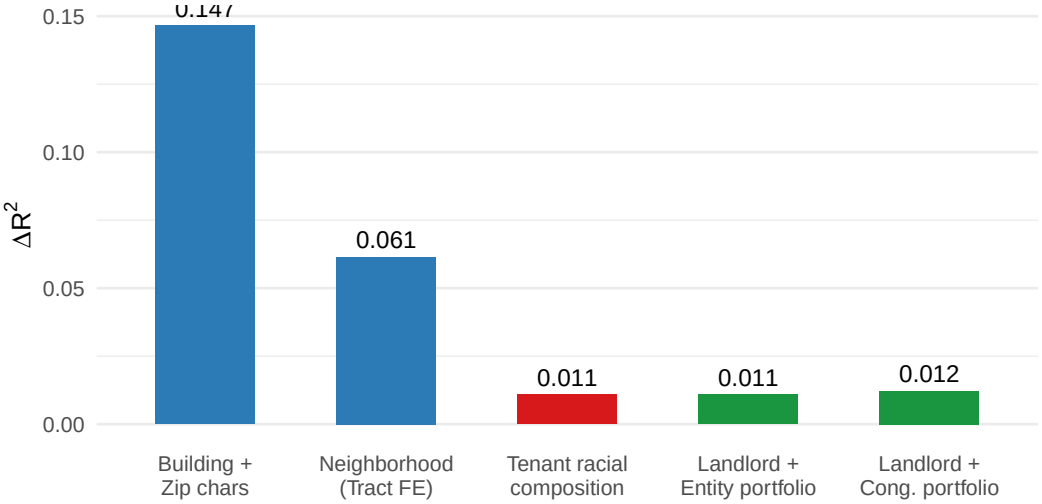

Figure 2: Incremental R-squared by component group. Outcome: EB filing rate. Weighted by unit count. Portfolio bars show delta from M3 for entity (M4) and conglomerate (M5) portfolio measures.

3.4 A3: OLS Coefficient Table (M1–M5)

This table reorganizes the owner-type comparison to match the bridge plot: first landlord characteristics only, then adding neighborhood controls, then building controls on the same common conglomerate-portfolio sample. Neighborhood controls now include both mean Black share and mean female share. Entries after the first two columns are owner-type coefficients relative to person owners, with clustered standard errors in parentheses.

Table 8: Owner-Type Bridge Regressions

Dependent Variable:	filing_rate_eb_pre_covid		
Model:	(1)	(2)	(3)
Variables			
Constant	0.0417***		
	(0.0018)		
For-profit corporate owner	0.0474***	0.0507***	0.0204***
	(0.0067)	(0.0062)	(0.0049)
Nonprofit owner	0.0430***	0.0527***	0.0060
	(0.0140)	(0.0128)	(0.0151)
PHA owner	-0.0616***	-0.0537***	-0.0991***
	(0.0143)	(0.0106)	(0.0130)
Religious owner	0.0085	0.0010	-0.0226
	(0.0259)	(0.0270)	(0.0233)
Trust owner	0.0328**	0.0397***	0.0210
	(0.0166)	(0.0149)	(0.0142)
Local government owner	-0.0377***	-0.0211	-0.0161
	(0.0135)	(0.0135)	(0.0136)
Federal government owner	-0.0674***	-0.1335***	-0.1049**
	(0.0127)	(0.0444)	(0.0454)
Financial intermediary owner	0.0247	0.0216	0.0094
	(0.0304)	(0.0240)	(0.0144)
Conglomerate portfolio: 2-4 properties	0.0056	0.0033	-0.0010
	(0.0067)	(0.0063)	(0.0055)
Conglomerate portfolio: 5-19 properties	0.0083	0.0023	-0.0022
	(0.0067)	(0.0051)	(0.0042)
Conglomerate portfolio: 20+ properties	0.0316**	0.0271**	0.0219**
	(0.0145)	(0.0110)	(0.0104)
Mean Black share		0.0463***	0.0351***
		(0.0053)	(0.0050)
Mean Female share		0.0097***	0.0017
		(0.0025)	(0.0023)
2-5 units			0.0011
			(0.0019)
21-50 units			0.0237***
			(0.0091)
51+ units			0.0202*
			(0.0103)
6-20 units			-0.0035
			(0.0062)
Rental-share correction			0.0527***
			(0.0035)
Fixed-effects			
Tract		Yes	Yes
Building type			Yes
Year built decade			Yes
Quality grade			Yes
Fit statistics			
Observations	172,871	172,870	172,870
R^2	0.03558	0.15841	0.23210
Adjusted R^2	0.03551	0.15648	0.23012

Clustered (Tract) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

3.5 A4: Market Concentration in ZIP x Filing-Rate Markets

To mirror the consolidation analysis from the other project, I construct a landlord market-year panel directly from the ownership panel. A **market** is defined as:

[market = ZIP code $\times 1\{\text{building EB filing rate} > 0.05\}$.]

This creates two submarkets within each ZIP: a higher-filing segment and a lower-filing segment. Within each market-year, landlord concentration is measured using conglomerate-level unit shares, so linked LLCs count as one owner. The main concentration metric is unit-based HHI on the standard 0–10,000 scale.

Table 9: ZIP x filing-rate market concentration summary in 2011 and 2019.

Segment	Year	Markets	Units	Mean HHI	Top-1 share (%)	Top-3 share (%)	Corp. unit share (%)	Median owners
Lower-filing ZIP segment	2011	50	248,916	113	6.2	10.8	35.9	1916
Lower-filing ZIP segment	2019	51	272,511	112	5.9	10.6	40.0	1887
Lower-filing ZIP segment	2011	52	97,893	203	7.2	14.9	43.3	518
Lower-filing ZIP segment	2019	53	101,381	209	7.2	15.5	48.5	496

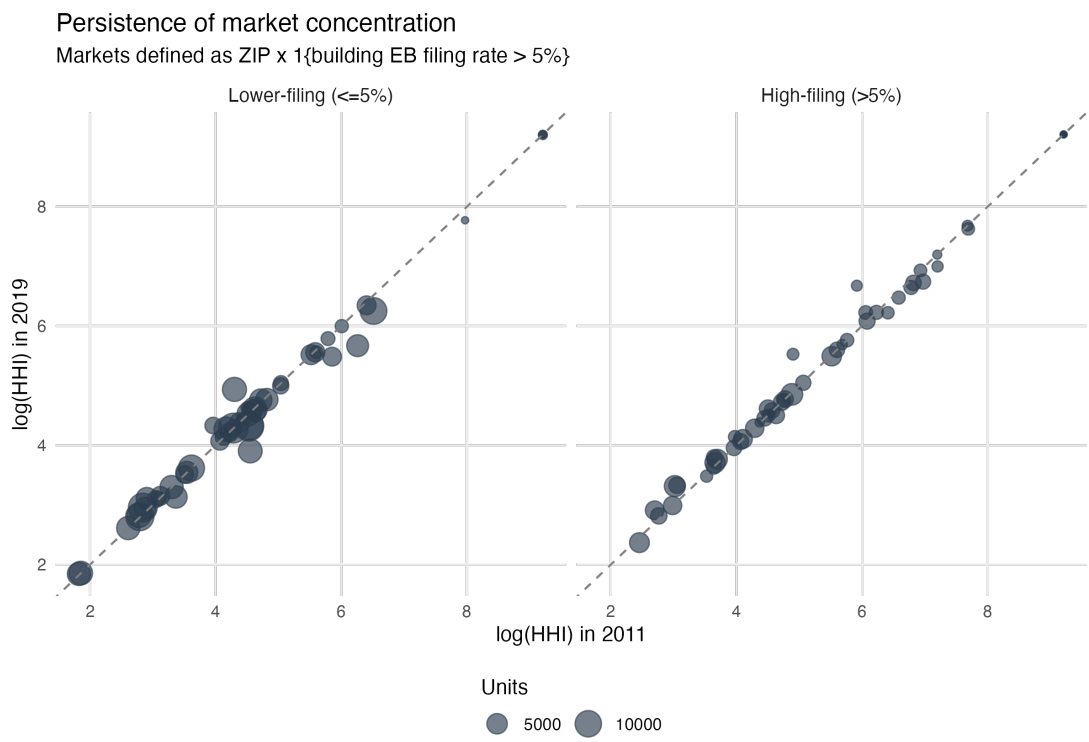

Figure 3: Persistence of log(HHI) between the first and last sample years. Each point is a ZIP x filing-rate market, sized by market units.

Across the sample, the higher-filing ZIP segments are consistently more concentrated and more corporate-owned than the lower-filing segments. The persistence plot shows that this is not just cross-sectional noise: markets that start concentrated generally stay concentrated through the end of the sample.

3.6 A5: Market-Year Rent Regressions

These regressions parallel the old consolidation script, but use the Philadelphia building panel. The outcome is the unit-weighted mean of building `log_med_rent` within each ZIP x filing-rate market-year.

Table 10: Market Concentration and Rent in ZIP x Filing-Rate Markets

Dependent Variable:	mean_log_rent				
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Log HHI	0.0250	-0.0051	-0.0259	-0.0280	0.0069
	(0.0270)	(0.0376)	(0.0818)	(0.0829)	(0.0674)
Corporate unit share (%)		0.0031	0.0074***	0.0074***	0.0060**
		(0.0029)	(0.0026)	(0.0026)	(0.0025)
Mean log income				-0.0190	-0.0365
				(0.0279)	(0.0281)
Log market units					0.2290**
					(0.1103)
Fixed-effects					
Year	Yes	Yes	Yes	Yes	Yes
market_id			Yes	Yes	Yes
Fit statistics					
Observations	869	869	865	860	860
R^2	0.07468	0.08424	0.90592	0.90597	0.90675
Adjusted R^2	0.06499	0.07357	0.89248	0.89245	0.89320

Clustered (market_cluster) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

The identifying variation in columns (3)–(5) is within market over time, after absorbing market and year fixed effects. In other words, these columns ask whether a given ZIP x filing-rate segment becomes more concentrated in years when it also becomes more expensive.

3.7 A6: Alternative Market Definitions

Two easy robustness checks use alternative market definitions: (i) census tracts, and (ii) ZIP code crossed with building-size bins `1`, `2--4`, `5--49`, and `50+` units.

Table 11: Tract-market concentration summary in the first and last sample years.

Year	Markets	Units	Mean HHI	Top-1 share (%)	Top-3 share (%)	Corp. unit share (%)	Median owners
2011	384	346,775	375	11.2	20.6	38.0	400
2019	384	373,854	371	11.1	20.6	42.3	411

Table 12: ZIP x property-size market concentration summary in the first and last sample years.

Size bin	Year	Markets	Units	Mean HHI	Top-1 share (%)	Top-3 share (%)	Corp. unit share (%)	Median owners
1	2011	52	138,190	55	3.7	5.3	23.7	1868
1	2019	52	140,435	54	3.7	5.3	26.6	1847
2-4	2011	48	89,766	43	3.2	5.1	26.4	676
2-4	2019	50	94,030	40	3.2	5.2	30.5	654
5-49	2011	49	67,264	276	7.7	15.9	57.7	63
5-49	2019	49	78,078	231	6.9	14.5	60.2	63
50+	2011	47	51,589	1185	19.4	43.1	70.8	8
50+	2019	48	61,349	1085	18.4	39.5	73.5	8

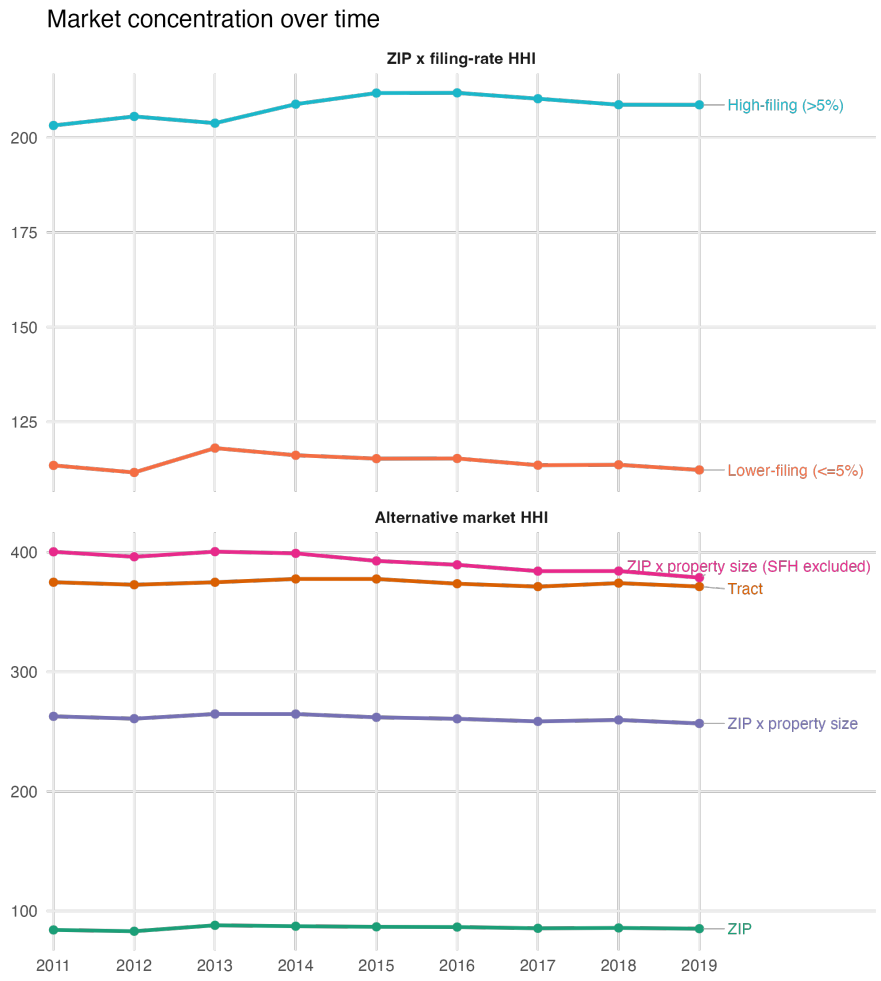

Figure 4: HHI over time in ZIP x filing-rate markets and alternative market definitions. End-of-line labels use the plotted series names.

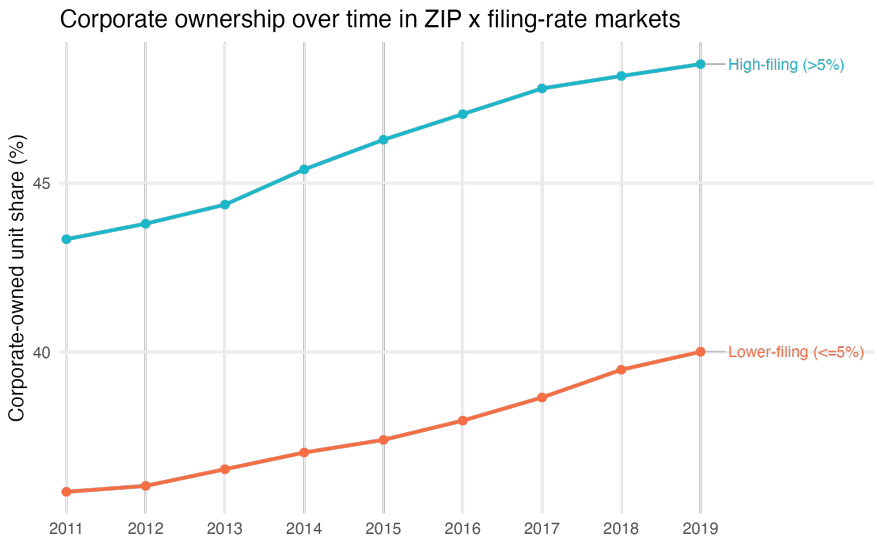

Figure 5: Corporate-owned unit share over time in ZIP x filing-rate markets. End-of-line labels use the plotted series names.

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4 Part B: Case-Level Back-Rent Analysis

4.1 B0: Sample

Table 13: Case-level sample construction

Step	Value
Eviction cases 2011–2019 (before commercial filter)	207,244
Commercial cases dropped (defendant #1 is a business entity)	5,669
Non-commercial cases retained	201,575
After rent/months filters (ongoing > 0, months >= 1)	163,644
After PID linkage (unique-parcel match)	153,746 (76.3%)
After rent winsorization (ongoing 300–6000, total 300–15000)	152,733
Entity ID (building owner) linked	95.5%
Plaintiff in name lookup	29.1%
Plaintiff is building owner (exact entity match)	15.8%
— Plaintiff type breakdown (winsorized sample) —	
For-profit corporate plaintiff	51.5% (78,615)
Philadelphia Housing Authority (PHA) plaintiff	4.1% (6,317)
Nonprofit / affordable-housing plaintiff	2.3% (3,518)
Individual plaintiff	42.1% (64,283)
— Owner type breakdown —	
Person owner	48.1% (73,467)
For-profit corp owner	46.6% (71,186)
PHA owner	3.6% (5,560)
Nonprofit owner	1.7% (2,520)

Only 15.8% of cases have a plaintiff matching the building owner entity exactly, confirming that management companies file the majority of evictions.

Note on PHA plaintiff cases: 6,317 PHA plaintiff cases survive the rent filter (4.1% of the winsorized sample). PHA income-based rents have a median of $221/month and roughly 78% fall below the $300 rent floor; the surviving cases represent above-poverty-line PHA tenants in higher-rent units.

Note on PHA owner type: 5,560 cases are in PHA-owned buildings (3.6%). The PHA-owned flag uses entity-ID matching through the name-lookup table, so it identifies the subset of cases where the building is formally owned by the Housing Authority regardless of who filed.

4.2 B1: Descriptive statistics

Table 14: Back-rent descriptives by landlord/plaintiff type and landlord scale (winsorized sample). Mean cong. units = mean total units across the conglomerate portfolio in the case year.

Category	N	Med. months	Mean rent	Mean cong. units	Pct. win
By plaintiff/owner type					
Non-fp-corp plaintiff	73,978	2.10	$730	732	5.3%
For-profit corp plaintiff	79,999	2.00	$838	232	2.0%
Non-PHA plaintiff	147,651	2.00	$803	143	3.7%
PHA plaintiff	6,326	1.72	$391	8025	1.8%
Non-nonprofit plaint.	150,437	2.00	$788	485	3.6%
Nonprofit plaintiff	3,540	2.12	$716	275	1.3%
Person owner	64,784	2.14	$765	15	5.6%
Corporate owner	81,981	2.00	$794	848	2.1%
Non-PHA owner	148,375	2.00	$800	103	3.6%
PHA owner	5,602	1.79	$408	9987	1.7%
Mgmt co. filer	125,000	2.00	$792	200	4.0%
Owner-filer	28,977	2.00	$761	1619	1.9%
High-filing bldg	80,721	2.00	$795	188	2.6%
Lower-filing bldg	61,874	2.19	$765	843	5.0%
By landlord scale (units outside this building)					
Scale: single-property	71,764	2.01	$816	33	4.6%
Scale: small landlord	24,051	2.09	$736	8	4.4%
Scale: large landlord	30,710	2.00	$761	2090	1.5%
Scale: medium landlord	15,708	2.10	$740	38	2.5%

4.3 B2: Months-of-Arrears Regressions

Outcome: log(months) of back rent at filing. Models use `plaintiff_type` (Individual = reference) and `owner_type` (Person = reference) as categorical variables rather than separate binary dummies, which keeps the reference group explicit and tests all plaintiff/owner types jointly. All specs include year × tract fixed effects. Specs B2–B4 add `total_units_conglomerate_yr_net_parcel_bin` — a four-category measure of the landlord's portfolio units in *other* buildings (i.e. the conglomerate total excluding the evicted parcel itself): single-property, small ($ $10units), medium(10 - -49units), large(50+$ units). This isolates scale effects from building-level unit count already controlled by `num_units_bin`. Specs B3–B4 further add building type and unit-size controls. Spec B5

is the back-rent elasticity model with the same landlord scale control. Standard errors clustered at the building (PID) level.

Key results:

- **PHA plaintiff** files fastest: coefficient ≈ -0.41 (B1) to -0.48 (B4), stable and large across all specifications. The effect is stronger than previously estimated, now that the sample includes single-unit rentals (PHA rarely evicts single-family tenants, sharpening the contrast).
- **Nonprofit plaintiff**: $\approx -0.14^{***}$ (B1), persisting at $\approx -0.10^{**}$ in B3. Nonprofits enforce noticeably faster than individual landlords even after controlling for building type and neighborhood.
- **For-profit corporate plaintiff**: $\approx -0.09^{***}$ (B1), shrinking to $\approx -0.02^{*}$ ($p = 0.036$) in B3 with full controls. The effect is now statistically significant — for-profit corporations file for modestly less arrears than individuals after conditioning on building characteristics and neighborhood.
- **Owner type** (Person = reference): For-profit corp and Nonprofit owners are indistinguishable from person owners ($p > 0.15$ in B3/B4). **PHA-owned buildings** have significantly *more* months of arrears at filing ($+0.09^{*}$, $p = 0.03$ in B3): tenants in PHA-owned buildings accumulate more debt before eviction is filed, consistent with PHA's more deliberate enforcement process.
- **High-filing buildings** evict sooner: main effect -0.12 (B3) to -0.18 (B4), highly significant throughout.
- **B4 interactions** (plaintiff type $\times$ high-filer building): All three corporate plaintiff types show significant positive interactions with the high-filer indicator (for-profit corp: $+0.085^{***}$; nonprofit: $+0.182^{***}$; PHA: $+0.326^{***}$). This means that in high-filer buildings, corporate and institutional plaintiffs file for *more* arrears relative to individual landlords in the same buildings — the speed advantage of institutional filers is attenuated in buildings where eviction is already a routinized practice.
- **Plaintiff-is-owner** near zero throughout ($p > 0.3$), consistent with management companies acting as agents rather than behaving differently from owner-filers.

4.4 B3: Back-Rent Elasticity

Outcome: log(total rent). Tests whether the back-rent amount scales proportionally with monthly rent ($\hat{\beta} = 1$) and how landlord type shifts the relationship.

The `log(ongoing rent)` elasticity is approximately **0.71** (significantly below 1 at any conventional level). This means higher-rent units are filed for proportionally *less* total back rent — consistent with landlords filing sooner at higher rent levels, possibly because vacancy and legal costs loom larger relative to each month of arrears. Among plaintiff types, only PHA shows a significant total-rent discount (-0.32^{***}), partly mechanical because surviving PHA cases are already above the \$300 rent floor and PHA policy triggers filing after one missed payment.

4.5 B4: Racial Disparity Regressions

These specifications use the imputed case-level defendant race file (`race_imputed_case_sample`) merged onto the winsorized case sample. The main race regressor is the **case-level Black probability**, not the building racial composition. All columns in Table 16 drop **PHA plaintiff cases**

Table 15: Back-Rent Months of Arrears Regressions

Dependent Variable:	log_months				
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
For-profit corporate plaintiff	-0.0811***	-0.0447***	-0.0186*	-0.0407***	-0.0064
	(0.0075)	(0.0109)	(0.0105)	(0.0125)	(0.0082)
Nonprofit plaintiff	-0.1427***	-0.0590	-0.0733*	-0.2221***	-0.0707***
	(0.0174)	(0.0372)	(0.0375)	(0.0387)	(0.0230)
PHA plaintiff	-0.4846***	-0.5214***	-0.5145***	-0.5468***	-0.4566***
	(0.0185)	(0.0410)	(0.0398)	(0.0396)	(0.0250)
Log ongoing rent	-0.2499***	-0.2334***	-0.2598***	-0.2629***	-0.2593***
	(0.0086)	(0.0086)	(0.0088)	(0.0088)	(0.0087)
For-profit corporate owner		-0.0047	0.0355***	0.0338***	
		(0.0116)	(0.0118)	(0.0117)	
Nonprofit owner		-0.0121	0.0177	0.0051	
		(0.0367)	(0.0360)	(0.0350)	
PHA owner		0.0364	0.0784*	0.0797*	
		(0.0471)	(0.0453)	(0.0448)	
Religious owner		-0.2049**	-0.1446*	-0.1491*	
		(0.0867)	(0.0810)	(0.0818)	
Trust owner		-0.0886**	-0.0577	-0.0570	
		(0.0381)	(0.0371)	(0.0376)	
Local government owner		-0.0157	0.0710	0.0785	
		(0.0750)	(0.1092)	(0.1095)	
Financial intermediary owner		-0.0838	-0.0619	-0.0623	
		(0.0512)	(0.0454)	(0.0461)	
High-filer building		-0.1485***	-0.1268***	-0.1508***	-0.1261***
		(0.0068)	(0.0070)	(0.0078)	(0.0070)
Other conglomerate units: small landlord		-0.0048	-0.0349***	-0.0336***	
		(0.0089)	(0.0090)	(0.0090)	
Other conglomerate units: medium landlord		0.0262**	-0.0080	-0.0061	
		(0.0120)	(0.0134)	(0.0133)	
Other conglomerate units: large landlord		-0.0333**	-0.0414***	-0.0398***	
		(0.0138)	(0.0144)	(0.0143)	
Plaintiff matches owner			-0.0162	-0.0140	
			(0.0109)	(0.0109)	
Building Black share			0.0012	0.0019	0.0009
			(0.0118)	(0.0117)	(0.0118)
2-5 units			-0.0703***	-0.0699***	-0.0706***
			(0.0102)	(0.0102)	(0.0102)
21-50 units			-0.1562***	-0.1520***	-0.1605***
			(0.0248)	(0.0247)	(0.0245)
51+ units			-0.2494***	-0.2435***	-0.2446***
			(0.0268)	(0.0267)	(0.0267)
6-20 units			-0.1013***	-0.0975***	-0.0982***
			(0.0236)	(0.0236)	(0.0235)
For-profit plaintiff x high-filer building				0.0407***	
				(0.0146)	
Nonprofit plaintiff x high-filer building				0.2108***	
				(0.0359)	
PHA plaintiff x high-filer building				0.3314***	
				(0.0432)	
Log conglomerate portfolio size					-0.0046*
					(0.0024)
Fixed-effects					
Year	Yes	Yes	Yes	Yes	Yes
Tract	Yes	Yes	Yes	Yes	Yes
Building type			Yes	Yes	Yes
Fit statistics					
Observations	142,649	142,167	118,264	118,264	118,522
R^2	0.11988	0.13116	0.14981	0.15076	0.14973
Adjusted R^2	0.11746	0.12870	0.14683	0.14776	0.14682

Clustered (PID) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 16: Back-Rent Elasticity Regression

Dependent Variable:	Log total back rent
Model:	(1)
Variables	
Log ongoing rent	0.7393***
	(0.0087)
For-profit corporate plaintiff	-0.0202**
	(0.0102)
Nonprofit plaintiff	-0.0703*
	(0.0378)
PHA plaintiff	-0.5187***
	(0.0394)
For-profit corporate owner	0.0317***
	(0.0116)
Nonprofit owner	0.0165
	(0.0365)
PHA owner	0.1103**
	(0.0501)
Religious owner	-0.1463*
	(0.0822)
Trust owner	-0.0575
	(0.0369)
Local government owner	0.0899
	(0.1098)
Financial intermediary owner	-0.0599
	(0.0458)
High-filer building	-0.1262***
	(0.0070)
Log conglomerate portfolio size	-0.0086***
	(0.0029)
Building Black share	0.0015
	(0.0117)
Fixed-effects	
Year	Yes
Unit-count bin	Yes
Tract	Yes
Building type	Yes
Fit statistics	
Observations	118,522
R^2	0.36508
Adjusted R^2	0.36287

Clustered (PID) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

and PHA-owned parcels. Column (1) includes only `case_p_black` with **no** year, tract, or building-type fixed effects. Column (2) adds the full fixed effects and controls. Column (3) adds the plaintiff-type interaction terms on top of column (2). Column (4) goes back to column (2) but removes tract fixed effects. The merged race file covers 153,392 cases in the winsorized sample; `Building Black share` remains in the table as a contextual control.

Table 17: Racial Disparity Regressions

Dependent Variable:	log_months			
Model:	(1)	(2)	(3)	(4)
Variables				
Constant	0.7371***			
	(0.0149)			
Case Black probability	0.1039***	-0.0036	-0.0181*	0.0040
	(0.0189)	(0.0077)	(0.0107)	(0.0087)
For-profit corporate plaintiff		-0.0285***	-0.0439***	-0.0441***
		(0.0104)	(0.0132)	(0.0154)
Nonprofit plaintiff		-0.0952***	-0.0992*	-0.0980**
		(0.0355)	(0.0561)	(0.0386)
For-profit corporate owner		-0.0006	-0.0009	0.0031
		(0.0112)	(0.0112)	(0.0188)
Nonprofit owner		-0.0275	-0.0278	0.0185
		(0.0342)	(0.0345)	(0.0393)
Religious owner		-0.1976**	-0.2019**	-0.1826*
		(0.0833)	(0.0832)	(0.1108)
Trust owner		-0.0915***	-0.0915**	-0.0931*
		(0.0353)	(0.0355)	(0.0515)
Local government owner		0.0394	0.0399	0.0231
		(0.1318)	(0.1320)	(0.1604)
Financial intermediary owner		-0.0958**	-0.0954**	-0.0575
		(0.0477)	(0.0478)	(0.0415)
Plaintiff matches owner		-0.0021	-0.0026	0.0022
		(0.0112)	(0.0112)	(0.0160)
Log ongoing rent		-0.3117***	-0.3116***	-0.3208***
		(0.0086)	(0.0086)	(0.0106)
Building Black share		-0.0167	-0.0142	0.0125
		(0.0120)	(0.0122)	(0.0120)
Case Black probability x for-profit plaintiff			0.0264*	
			(0.0136)	
Case Black probability x nonprofit plaintiff			0.0089	
			(0.0523)	
Fixed-effects				
Year		Yes	Yes	Yes
Tract		Yes	Yes	
Building type		Yes	Yes	Yes
Fit statistics				
Observations	113,119	113,118	113,118	113,119
R^2	0.00506	0.14517	0.14524	0.10863
Adjusted R^2	0.00505	0.14210	0.14216	0.10841

Clustered (PID) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 18: Key findings summary

Finding	Evidence
Neighborhood sorting is the dominant source of filing rate variation	ΔR^2(tract FE) $\approx$ 0.082 vs. ΔR^2(landlord) $\approx$ 0.009
Landlord type (owner type + entity portfolio) explains $\approx$0.9% of cross-sectional variation	M4 incremental R^2 from M3; robust to PPML
Conglomerate portfolio (M5) explains similar share to entity portfolio (M4)	M5 ΔR^2 from M3 comparable to M4; conglomerate portfolio in B5 reg.
PHA files fastest: 41–48% fewer log-months, highly stable across all specs	B1: -0.41^{***}; B4: -0.48^{***}; 6,317 cases survive \$300 rent floor
Nonprofits file faster; for-profit corp effect significant with full controls	Nonprofit B3: -0.10^{**}; for-profit corp B3: -0.02^{*}, $p = 0.036$
High-filing buildings evict sooner; corporate types partially offset this in high-filer bldgs	B4 interaction for-profit$\times$high-filer: $+0.085^{***}$; nonprofit: $+0.182^{***}$
PHA-owned buildings accumulate more arrears before filing ($+0.09^{*}$)	B3 owner_typePHA: $+0.090^{*}$, $p = 0.030$
Back-rent elasticity w.r.t. monthly rent = 0.71 (significantly below 1.0)	Elasticity $\approx$ 0.71; 95% CI excludes 1.0
Plaintiff = building owner in only 15.8% of cases	Name lookup exact match: 29.1%; entity match: 15.8%

5 Summary

5.1 Caveats

1. **Plaintiff lookup rate (29.1%):** Most eviction plaintiffs are management companies not present in RTT or OPA by exact name. Fuzzy-name matching (Phase 3, deferred) could substantially improve coverage.

2. **OPA fallback ownership (63.4% of PID-years):** Buildings with no RTT history inherit their 2024 OPA owner for all years, introducing attenuation bias if ownership changed during 2011–2019.

3. **Improved entity reconciliation (Phase 2/3):** As of the 2026-03-10 pipeline update, persons and trusts are grouped by (name, PID) rather than name alone. Phase 2 now uses name+address matching (not pure address sharing) to prevent registered-agent and government-hub explosion. Phase 3 adds a name-similarity gate preventing tax-exempt government conveyances from polluting private landlord conglomerates. These improvements make `portfolio_bin_conglomerate_yr` (M5/B5) a more reliable measure of true landlord scale than entity-level portfolio (M4).

4. **Racial disparity specs (B4):** Now active. The case-level defendant race file (`race_imputed_case_sample`) merged onto 153,392 cases in the winsorized sample. The current Table 16 specs drop PHA plaintiff cases and PHA-owned parcels, leaving 113,118–113,119 observations in the race-regression sample depending on the specification.